
NHL Zone Entries - Do teams that primarily control the puck into the zone generate more offensive efficiency than teams who dump and chase?

1. Introduction

Producing goals in the NHL often begins with the zone entry. A big debate is which strategy is more effective – controlled zone entries (“carry-ins”) vs dump-and-chase entries. At the player level, controlled entries lead to more shots and chances, however it is unclear whether this is the case at the team level over the course of an entire season.

This project examines whether NHL teams that rely more on controlled entries actually produce better offensive results once outside confounding factors such as shot volume, puck possession, shooting talent, and special teams ability are taken into account. Using team-season data from the 2022-23, 2023-24, and 2024-25 NHL seasons — including zone entry data from All Three Zones and all other data from Money puck and Evolving Hockey — this study evaluates whether a higher controlled entry rate causally improves a team’s offensive efficiency.

This question is important both analytically and strategically. If teams that carry the puck more consistently generate more expected goals or scoring chances, this would support tactical systems that prioritize transition speed, puck support, and controlled entries. If not, it may suggest that the observations at the player level do not scale to team level outcomes, perhaps due to roster construction, system differences, or defensive adjustments. The goal of this project is to isolate the independent effect of entry strategy on team offensive performance using statistical inference and regression modeling techniques.

2. Model Design and Data Overview

$$xGF60_{it} = \beta_0 + \beta_1 \text{ControlledEntryRate}_{it} + \beta_2 \text{FF60}_{it} + \beta_3 \text{CF60}_{it} + \beta_4 \text{ShootingPct}_{it} + \beta_5 \text{PPPct}_{it} + \beta_6 \text{PointsPct}_{it} + \gamma_t + \epsilon_{it}$$

where:

i = team

t = season

γ_t = season fixed effects

ϵ_{it} = error term

xGF60 = expected goals for per 60 minutes

ControlledEntryRate = carries60/total_entries60

FF60 = unblocked shot attempts per 60 minutes

CF60 = all shot attempts per 60 minutes

PPPct = powerplay%

Null Hypothesis ($H_0: \beta_1 = 0$): no relationship between controlled entry rate and xGF60

Alternative Hypothesis ($H_a: \beta_1 \neq 0$): a relationship exists between controlled entry rate and xGF60

Before coming to the final model above, I experimented with my variables by running multiple models to determine the most efficient way to answer my question.

The first decision I had to make was whether to use xGF60 or GF60 as the outcome variable.

Looking at Table 1, you will see that I began by using GF60 as my base model to start. This model captures the actual production of each team, however

this contains a lot of noise. For example, if a team got “unlucky” or was “too lucky”, this value might be skewed. Having a lot of noise in the outcome variable was suspect and something that I did not want. In Table 1, you will see that the base model with GF60 as the outcome variable has a higher r-squared, and lower AIC + SIC when compared to Alt Model 3 (the one that I ultimately used). These numbers would usually suggest that I should use GF60 as the outcome variable. However, Model 3 still demonstrated a strong R^2 and low AIC/SIC. This, combined with the noise mentioned earlier, led me to use xGF60 as my outcome variable (Alt Model 3), since this measures the true offensive efficiency of each team. The xGF60 values for each team came from Evolving Hockey, who use a formula to generate them.

My primary independent variable had to measure entry strategy, which is why I used ControlledEntryRate in this case. This value was calculated by taking each team’s “Carries60” and dividing it by their “TotalEntries60”, to get the percentage of zone entries that this team uses a controlled entry. The alternative to carrying the puck into the zone is dumping the puck. To avoid perfect multicollinearity, I did not include “DumpEntryRate”, since you can simply subtract the ControlledEntryRate from 1 to get that value.

The rest of my variables are controls which serve the purpose of isolating the effect of ControlledEntryRate on xGF60. Without the following controls, there would be omitted variable bias and β_1 would be biased:

FF60 (Fenwick For per 60) measures shot volume and scoring chances. Higher volume teams naturally produce more xGF. CF60 (Corsi For per 60) measures puck possession and territorial control. Teams controlling the puck more will have higher total entries and tend to carry the puck more (controlled entry = possession). ShootingPct (goals/shots on goal) reflects finishing talent and luck. Teams with better shooters will generally have higher xGF60. Powerplays (PP%) are a crucial event that must be controlled for. First of all, teams almost always carry the puck into the zone on the powerplay because they have an extra skater on the ice so it is a lot less difficult. Second of all, powerplays also lead to higher xG, further proving that this is a crucial control to include in the model. Finally, PointsPct (standings points) is a control for team strength.

Comparison of Model Fit Statistics					
Model	R-Squared	Adj. R-Squared	AIC	SIC (BIC)	Observations
Base Model - GF/60	0.959	0.955	-233.1	-207.2	99
Alt Model 1 - Chances/60	0.508	0.464	255.0	280.9	99
Alt Model 2 - Parsimonious (robust check)	0.836	0.825	-120.1	-99.3	99
Alt Model 3 - xGF/60 (final)	0.842	0.828	-120.0	-94.1	99
Note: Lower AIC/SIC and Higher R^2 indicate better fit.					

Table 1: Comparing Models

Without this, β_1 might capture the fact that team i is simply a better team. Stronger teams will attack more, sustain zone pressure, and often have more controlled zone entries.

Summary Statistics for Zone Entry Efficiency Model				
Variable	Mean	SD	Min	Max
xGF_60	3.106	0.302	2.470	3.800
controlled_entry_rate	0.477	0.038	0.410	0.570
FF_60	42.992	3.510	34.880	51.870
CF_60	58.968	4.946	46.390	71.260
shooting_pct	0.103	0.009	0.085	0.125
pp_pct	0.212	0.039	0.118	0.324
points_pct	0.554	0.102	0.287	0.823

Table 2: Summary Statistics for Model

among the teams in the dataset. Points percentage having a mean above .500 is due to the NHL's point system, where an overtime loss rewards a team with 1 point. This leads to some games distributing 3 points to the two teams competing, rather than just 2 points to the winning team and 0 to the losing team. Each game is guaranteed to be at least a 2 point game, however the occurrence of overtime 3 point games drives the average points percentage slightly above .500.

3. Model Results and Diagnostics

Table 3 displays the regression results. Interestingly, the main independent variable, `ControlledEntryRate`, is not statistically significant. This suggests that *how* you enter the zone might not matter as much as what you do once you actually enter the zone. Simply increasing the *number* of controlled zone entries will not create an increase in xGF. This means that a zone entry decision should be made based on the defensive alignment. Forcing a controlled entry against a

Table 2 shows the summary statistics for each of these variables. Note that despite being shown in decimal form, `controlled_entry_rate`, `shooting_pct`, `pp_pct`, and `points_pct` are all percentages and should be read as such. The dependent variable xGF60 does not have a lot of variation, meaning most teams average right around 3 xGF60. On the other hand, the possession and shot volume variables have the most variation, with wide ranges

OLS Regression Results: Factors Affecting Expected Goals For (xGF/60)		
Term	Estimate	p-value
Intercept	-0.899**	0.006
Controlled Entry Rate	0.481	0.197
Fenwick For (FF/60)	0.047*	0.014
Corsi For (CF/60)	0.02	0.135
Shooting Percentage	4.976*	0.019
Power Play Percentage	1.534***	<0.001
Points Percentage	-0.243	0.267
R ² = 0.842 Adjusted R ² = 0.828 Observations (N) = 99 Stars: *** p<0.001, ** p<0.01, * p<0.05		

Table 3: Regression Results

tight defense might result in a turnover at the blue line, which eliminates the theoretical “benefit” of the carry in.

Looking at the other variables, we see that Fenwick For is statistically significant, however Corsi For is not. This is also interesting, since these variables are similar and usually looked at together, so we would expect them to either both be significant or both be insignificant.

Why might this be the case?

Fenwick measures unblocked shots, while Corsi includes measures for all shot attempts (including blocked ones). This result suggests that getting shots through traffic (unblocked) is a significant driver of xGF, whereas just flinging pucks that get blocked (Corsi) is not. The similarities between Fenwick For and Corsi For that I just mentioned are proven to be true

Multicollinearity Check (VIF)	
Variance Inflation Factor by Variable	
Variable	VIF Score
Controlled Entry Rate	1.22
Fenwick For (FF/60)	26.79
Corsi For (CF/60)	26.14
Shooting Percentage	2.19
Power Play Percentage	1.66
Points Percentage	3.09
Season	2.57
Note: VIF > 5 indicates potential multicollinearity. VIF > 10 indicates severe multicollinearity.	

Table 4: Multicollinearity Check (VIF)

when examining the VIF score for each variable in the model in Table 4. The ridiculously high VIF values of ~26 for these variables are expected given their similarity. The decision to keep both FF and CF in the model despite them having alarming VIF values is due to the conclusion that can be drawn from the results. Without including both, I would not have been able to analyze shot quality vs quantity, which is a big conclusion that can be drawn from this model. It is a good sign that all other variables in the model have VIF's below 4, this means that aside from Fenwick and Corsi, no other variables are highly correlated with one another.

To further analyze the model used, I ran a Breusch-Pagan test to test for Heteroscedasticity, as well as a Durbin-Watson test to test for Autocorrelation. Looking at

Model Assumption Checks			
Analysis of Residuals for Zone Entry Model			
Test	Statistic	P-Value	Interpretation
Heteroscedasticity (Breusch-Pagan)	10.33	0.242	Constant Variance (Good)
Autocorrelation (Durbin-Watson)	1.76	0.075	No Autocorrelation

Table 5: Model Diagnostics

Table 5, we see the results from these two tests, both of which show good results. For the Breusch-Pagan test, the null hypothesis states that the variance is constant, which is what we want. Given the insignificant p-value from the test

of 0.242, this means that we fail to reject H_0 and cannot prove that the variance is not constant. Similarly, for the Durbin-Watson test, the null hypothesis is that there is *no*

autocorrelation. Again, we see an insignificant p-value of 0.075, meaning that we cannot conclude that there is autocorrelation. These two diagnostic tests confirm that the model satisfies the assumptions of OLS regression.

4. Analysis of Results

The model performed very well, explaining 84.2% of the variation in Expected Goals (xGF/60). This high R-squared value is critical not just for explanation, but for inference; it confirms that the control variables accurately capture the primary drivers of offense, allowing us to isolate the specific impact of entry strategy.

Interestingly, simply entering the offensive zone with control did not matter as much as we thought. Controlled Entry Rate ($p = 0.197$) was not statistically significant in the model. This means that the type of zone entry does not significantly impact the production following the entry. The model suggests that even if a team were to drastically alter their system and increase their controlled entry rate by 10 percentage points (a massive tactical shift), the associated rise in xGF/60 would be insignificant and not statistically distinguishable from zero.

The rest of the results show that shot quality and special teams matter most in terms of creating xGF. Power Play Percentage ($p < 0.001$) was the strongest predictor, simply implying that teams with better power plays generate higher expected goals, as expected. Shooting Percentage ($p = 0.019$) and Fenwick For ($p = 0.014$) were also key drivers, meaning that finishing talent and getting unblocked shots on net are critical for offense. Also, while unblocked shots (Fenwick) were significant, total shot attempts (Corsi) were not. This suggests that just flinging the puck at the net doesn't help much if the shots are getting blocked; getting the puck through traffic is what drives expected goals.

To quantify the impact of the statistically significant variables, we can analyze the marginal effect of a one unit increase in each metric on Expected Goals (xGF/60), holding all else constant:

Shooting Percentage - This was the most sensitive driver of offense. An increase in team Shooting Percentage of just 1 percentage point (ex: improving from 10% to 11%) corresponds to an increase of roughly 0.05 xGF/60. Over the course of an 82 game season, improving shooting efficiency by just 1% would result in approximately 4.1 additional expected goals.

Fenwick For - Generating more unblocked shots reliably improves offensive output. For every 1 additional unblocked shot (Fenwick) per 60 minutes, xGF increases by 0.047. While small on a per shot basis, 5 more unblocked shots per game generates roughly 19 additional expected goals over a season.

Power Play Percentage - Special teams efficiency is also key. A 1 percentage point increase in Power Play conversion rate is associated with an increase of 0.015 xGF/60. While this effect appears smaller than shooting percentage, it is statistically significant

($p < 0.001$), confirming that capitalizing on the man advantage directly translates to overall expected production.

5. Conclusion

Overall, the insignificant p value for the main independent variable of ControlledEntryRate means that we fail to reject the null hypothesis and can not conclude that there is evidence that a higher rate of controlled entries independently drives higher Expected Goals compared to dump and chase strategies. This suggests that the method of entry (Carry vs. Dump) matters less than the outcome of the entry. Putting all of this together to answer the underlying question: Primarily “carrying” the puck into the zone does not statistically lead to higher generated offense.

Finally, it is critical to acknowledge the limitations imposed by the availability of public tracking data. This study relied on zone entry data from All Three Zones that was only available for three seasons (2022-2025), resulting in a sample size of 99 team-season observations. From a statistical inference perspective, this relatively small “N” limits the power of the model. With a small sample, standard errors are naturally larger, which makes it more difficult to reject the null hypothesis even if a true, underlying relationship exists.

Word Count (excluding title/headers, tables): 2,073